

# Coalition Formation with Logrolling: An Extension of Coleman’s Framework

*[Working title — draft manuscript]*

Summer 2019

## Note on this revised document

This is a revised version of the summer 2019 draft manuscript. Figures and makeshift tables from the original document have been replaced with generated visualisations. Passages identified as author comments, internal notes, suggestions, or to-do items are enclosed in amber boxes like this one.

A coalition may be formed when members of a larger collectivity anticipate that they can gain more from banding together than by operating as separate units — bonds bring benefits. Coalition theory is about understanding when and why coalitions are built or .

Coalitions are ubiquitous in social life and range from alliances between states to firms constructing cartels and political parties negotiating government formation. There are many classical accounts of coalitions, e.g. in Herodotus’ history of the Greco-Persian wars or Thucydides’ history of the Peloponnesian War. In *The Prince*, Machiavelli formulated maxims on coalition formation. And ideas of how realignments among coalition members can result either in a balance of power, which is at the heart of the analyses of the “Concert of Europe” throughout the 19th century, or result in one partner outmaneuvering all others to attain political dominance, such as in accounts of the Russian Revolution.

As for theory, the seminal modern work on coalition formation was von Neumann and Morgenstern’s *Theory of Games and Economic Behavior* (1967, 1947), which introduced new concepts and new modes of analysis. As Francis Bloch has stated: “Since the publication of *Theory of Games and Economic Behavior*, the study of endogenous formation of coalitions has been one of the most intriguing and challenging problems open to game theorists.” Von Neumann and Morgenstern’s classic spawned a rich, but quite technical literature in several disciplines, notably in economics and political science. Short, informative surveys have been provided by Siddhartha Bandyopadhyay and Kalyan Chatterjee

(2006), and Myrna Wooders and Frank H. Page (2013).

## 1 Riker's analysis

Within the field of political science, William G. Riker's *The Theory of Political Coalitions* (first published in 1962) has been very influential. He analyzes zero-sum games where  $n$  actors with perfect information can engage in transactions and side-payments are allowed. In such groups, assembling a coalition "is a process of forming a subgroup which, by the rules accepted by all members, can decide for the whole" (12).

In line with von Neumann and Morgenstern (1967, 1947), Riker distinguishes between two types of gains from a coalition.

The first is the payoff to the coalition as a whole, by other authors sometimes called the value or worth of a coalition. Since different possible coalitions can yield different payoffs, the set of payoffs to each possible coalition in a game can be characterized. This is called the characteristic function of the game. If one can specify and list the payoffs to the coalitions, it will be apparent that some coalitions are more profitable than others and "it can, provisionally, be supposed, *ceteris paribus*, the less profitable ones will not be considered by the players." (36)

The other type of gain is each player's own payoff, which Riker, like von Neumann and Morgenstern, calls an imputation. An imputation is a payoff vector, or "a list of the payment to each player in a given structure of coalitions." (37). This payoff vector essentially limits the coalitions a player will seriously consider and hence affect the process of coalition building. Realizable coalitions are those that rational players could be willing to enter.

In sum: What individual players get depends on which coalition they join. Hence there can be a tension between the overall payoff to a coalition and the spoils to its members — possibly a coalition with a high overall payoff can yield a lower payoff to a potential member than an alternative coalition, or a member may be excluded from a potential coalition if he reduces the payoffs to the other members of that coalition and the coalition without him is still realizable. Both make for minimizing the size of a coalition.

This "size principle" is one of Riker's main contributions: "In social situations similar to  $n$ -person, zero-sum games with side-payments, participants create coalitions just as large as they believe will ensure winning and no larger." (47) However, several winning coalitions may be possible yet without one of them trumping the others in terms of the payoffs. (46) Other insights from Riker's analysis are:

- The zero-sum assumption implies that what winners gain is equal to the sum of

what the losers lose. When a coalition includes everybody, the winners gain nothing simply because there are no losers. (39)

- In any coalition larger than the minimum, its members can increase the amount to be divided among them by ejecting one or more of their members. Since the ejection is presumably costless, in such a game only minimum winning coalitions can be expected to occur. (42)
- The winning coalition has a strong incentive to prevent the formation of an alternative coalition. (42)
- A highly significant assumption of the model is that the members have control over additional entries into their coalition. If they have no such control, all losers could invariably join the winners and thereby produce a valueless coalition of the whole and nullify the winners' victory. (39)

Riker also develops a dynamic model of coalition formation. Here his key idea is that from the “coalition of the whole”, a number of subsets — proto-coalitions — can potentially be formed. They can have different value to the potential members, and the members can influence the decision of the whole by their weight — e.g. by the fraction of votes they control.

Moreover, Riker assumes that the process of coalition formation is one of bargaining. For example, in a legislature the members can trade votes or promises to vote (logrolling and reciprocal concessions), which may have a value in bilateral transactions as well as for a coalition as a whole. Or they may use side payments — i.e. providing some benefit over and beyond the transaction itself to induce the recipient to enter the transaction.

Riker emphasized that such side payments are valuable, regardless of whether they are in terms of material things such as money or take the form of promises on policy. To say that they have value is to say that some people want them badly enough to give up something else to obtain them. (106)

Potential affiliates can change the composition of proto-coalitions and thereby the relative weight of members of the coalition. Entry or exit also changes the concerns they bring to the table, how spoils are split, and hence the value of a proto-coalition to members.

The gain to the coalition as a whole and what accrues to each member, is a function of the degree of correspondence or opposition of their attitudes towards issues, i.e. being in favor or against passing them, as well as the number of potential members. “Excess members of a coalition both cost something to acquire and lessen its gains. And this is to say that the scarcity of side-payments prohibits their use in an extravagant fashion.” (107)

What can be the object of bargaining is not just how the coalition's votes are to be cast — for or against — different bills, but also what is to be the content of bills, i.e. what specific items are to be included or excluded. However, policy changes are valuable only when they are controversial, only when they involve the satisfaction of one interest at the expense of another. In a sense the value of a policy is a function of a kind of scarcity, namely a scarcity of beneficiaries. (106)

Riker's *Theory of Political Coalitions* is one of the most quoted sources among students of legislatures, alliances, and the like — in July 2019, Google Scholar listed 4577 citations. However, his theory has been criticized on several accounts. Robert Butterworth (1971) argued that the size principle is in error. Instead he suggested a “maximum positive gainers” and demonstrated that the winning coalitions may not be of the minimum winning size. Kenneth Shepsle (1974) introduced alternative solution concepts, “the von Neumann–Morgenstern stable set and the Aumann–Maschler bargaining set, and the implications they hold for the size of minimum winning coalitions.” (Shepsle 1974, p. 505).

Yet questions remain about Riker's size principle. Will minimum winning coalitions be unique in the sense that they will dominate all others? Are they stable, in the sense that a member will not be persuaded to leave it for a coalition that may yield a higher payoff for that coalition, even if it is not a minimum winning one? What other factors than the weight or share of a total votes of a potential member can enter such calculations? We return to these questions below.

## 2 Coleman's model: The basics

James S. Coleman has had strong influence on the theory of collectivity decisions. He published his magnum opus in 1990, *Foundations of Social Theory*. Two decades later, in July 2019 it had been cited 36 503 times according to Google Scholar. In 1970 he published an article specifically on coalition theory, “The Benefits of Coalition.” Compared both to Riker's book and his own *Foundations*, it has been cited only sparsely — according to Google Scholar in July 2019 only 27 times. Yet it is very suggestive for coalition research.

In his paper, Coleman examined two situations. Members of a collectivity vote on each collectivity action in isolation. Then coalitions may form either on a single decision or a set of decisions. In the second situation the members can engage in logrolling or vote trading across decisions.

The key questions addressed in “The Benefits of Coalition” were: “Under what conditions is it in my interest to join a coalition in a system of voting, and how much benefit does it bring me?” (45) Coleman sees coalition formation as a two-stage process: Instead of all members of a group acting independently in collectivity decisions, a subset (i.e.

coalition) first engages in a collectivity decision among themselves, and, having decided, will use their resources as a single unit. However, presumably, they would only join such a coalition if it were more beneficial to them than acting within the larger collectivity alone.

We will use the term *collectivity* for the ensemble of actors under consideration and *(proto)coalition* for subunits that (can) enter into a collaboration. So, under which conditions is it to an individual member of the collectivity to join a coalition, and how could he know? What are the effects on a collectivity decision of a two-stage representative democracy, where the collectivity is partitioned into sub-units or constituencies? And, what is the optimum size of a coalition for an individual to join in order to maximize control over issues, given a particular distribution of interests in the collectivity relative to one's own? In all these problems, the individual's decision depends on two considerations: on the one hand, the increased power the collectivity brings when it does act in his interest, and on the other, the loss he suffers when it fails to act in his interest, or acts in direct opposition to his interest.

It is worthwhile to note that a key source of inspiration and insight to answer such questions, as well as for his general conceptualization, Coleman derived from a simulation game, *Legislature*, which he had constructed in the early 1960s (Coleman 1964, 1969). In this game the players — “legislators” — may favor or oppose the passing of a given bill — that is, they have a positive or negative attitude towards passing the bill.

However, their stance is more than just an attitude, it is also one of intensity or strength, which Coleman calls an actor's *interest* in the outcome of the vote on the issue. Legislators have varying interests in different bills, e.g. by strongly favoring or opposing some bills while being relatively indifferent to other issues.

At the outset each legislator or actor can affect outcomes by the way they cast their own votes in line with their own immediate interests. But since each actor is interested in the outcomes, he is also interested in how other actors cast their votes. Hence each can affect the outcomes by more than his own vote if he can get other actors to vote in his interest.

Actors are therefore motivated to trade promises to vote for or against particular bills — i.e. to engage in logrolling: Each would promise to vote for or against issues that interest them less in return for a like promise on issues that interest them more. Indeed, each may promise to vote for a bill he is against when it is of little significance in return for a vote of great import. The value of such logrolling can be quantified. To do so, Coleman constructed a framework to analyze general systems with a perfect market for vote exchanges.

In its most basic form, a system of  $n$  actors engaging in bargaining or logrolling to come

to a collectivity decision on  $m$  issues can be represented by two matrices.

The first, the *interest matrix*, maps the actors' interests in the outcomes of the joint vote on a bill or issue (generally called an event), where the symbol  $x_{ji}$  represents the strength of actor  $j$ 's interest in event  $i$ . The strengths of these interests are calibrated to add to 1 across all events; hence one can immediately see an actor's interest in an event relative to his interest in other events. His attitude, positive or negative, towards an event is represented by a plus or minus sign in front of his interest. In other words, the direction of an actor's interest is distinguished from its strength.

Second, the *control matrix* maps the relative control an actor  $j$  initially has over an event  $i$ , denoted  $c_{ji}$ . In the model that control is calibrated to add to 1 across all actors; hence one can immediately see an actor's control over the outcome of an event relative to that of other actors.

This conceptualization means that interests are intrapersonal, control is interpersonal. What drives the system is the difference between what an actor controls and what he wants to control. Or put differently: what drives the system is the dissatisfaction of the actors with the initial status quo represented by the control matrix.

Hence the system is one of double contingency: The actors act in relation to each other, and the joint outcomes of the actions they take are interlinked. Thus, the actors are interdependent in a system with a personal profile of preferences as well as a social distribution of control or rights.

Because actors typically have distinct interests in and control over events, they may enter into transactions to improve on the status quo, giving up control over events that interest them less in exchange for control over events that interest them more. These transactions will move the system towards an equilibrium of final directed control of the actors over the events after transactions, which can then be summarized in a matrix, and depends directly on the given initial control and interest matrices (see appendix for precise mathematical formulation of this and the concepts introduced below). This final control expresses not only what amount of control an actor has gained or retains after transactions or trades, but also in which direction, for or against, it will be cast.

Coleman's basic framework can be used to derive other concepts, such as the overall power or resources of each actor relative to that of the others, the value of controlling an event relative to controlling other events, the power of a collectivity to act given the extent of opposed interests, etc.

There are several benefits to the use of this framework. First it is simple, both in terms of the basic assumptions as well as derived concepts and criteria for choice. Secondly, it is also easy to find intuitive correlates of these concepts, i.e. notions that real-life actors

entertain when bargaining about coalition formation. Thirdly, the basic model is in a sense the mirror image of a market where actors have full control over private goods which they barter; in the Coleman model they have partial control (votes) over public goods and bargain over how their votes shall be cast, since how an actor votes not only affects his own interest directly but also other actors indirectly through the externality of that vote for the interest of other actors.

To complete the framework, one more thing must be mentioned. In his conceptualization Coleman assumes a probabilistic decision rule. For example, if  $n_1$  of the actors vote in favor of an action, a probability mechanism is used to determine the outcome, such that it will pass with probability  $n_1/n$ . The upshot of vote trading is that the number voting for or against different bills will change.

With this conceptualization one can within the framework derive the probability of a positive outcome of a particular issue, given specific control and interest matrices. Furthermore, this derived value together with the directed interest each actor has in the outcome of the issues under consideration, can be used to quantify the value of a particular coalition to an actor: a coalition likely to vote in line with an actor's attitude and interest on different issues will have a high value for him, while a probability of voting against his interest will lower its value. Such calculations can then be used to compare the value of different potential (proto)coalitions to the actors.

Coalition theory becomes interesting when in a collectivity more than one feasible majority coalition can form. The question then becomes which will in fact see the light of day. There is also the possibility that a party left out of a majority coalition may be able to entice one of its members to leave it and form a new one. Indeed, there is also the possibility that bargaining enters a loop, in such a way that no stable coalition can form.

The number of possible coalitions, how many of them can be winning, whether a winning coalition will be stable in the sense that no member can be better off by joining another coalition of non-members, and whether a stable coalition is of the minimum winning size, are key questions. Further complexity may be added by adding or deleting issues, by changing the parties' or actors' interests, the numbers of votes that they have, and so on.

It is such complexities that make it desirable to have a theory that is fairly simple, both in terms of basic assumptions and derived concepts, as well as having criteria for choice among coalitions which are intuitive. These are precisely two of the advantages of building on Coleman's model. By adding a few concepts, it is possible to gain additional insights about not only the benefits of coalition, but also about the possible losses that outsiders may incur from a given coalition.

## Criteria for Choosing a Coalition

To make a choice between different possible coalitions, each actor is typically assumed to look out for his own interest. Moreover, we assume that an actor will wish to end up in the coalition that has the highest value for him. The situation *ex ante* is bargaining within the collectivity as a whole (or the coalition-of-all). Each actor operates as a singleton, so to speak, bargaining with all the other members. The situation is not one of a war of all against all, but one of bargaining all with all. The benefits the actors can obtain in this situation voting independently in a sense provides us with a baseline.

The question for each actor is whether he can improve on this situation by entering a coalition smaller than the collectivity of the whole. To answer this question, we must compare the outcome of each (smaller) coalition or proto-coalition for each actor. Hence, for each proto-coalition we calculate the expected realization of interest for each of its members. Then the question arises: is there one proto-coalition which ranks at the top for all the members of that proto-coalition? Furthermore: is that proto-coalition at the same time the minimal winning coalition, as Riker would have us expect? Or are several coalitions feasible? If so, are they stable, in the sense that no member could be enticed out of that coalition with the promise of a winning coalition that might yield a higher payoff for him?

For each coalition we may ask not only what is its value to its members, but in addition: what is its value, positive or negative, to a non-member? The answer, of course, would be an expression of the external effects of that coalition's members acting in their own interests for its non-member(s). So for each proto-coalition we can calculate its (positive or negative) value to the opposition.

If there are only three parties in the collectivity, and they have unequal votes or control, is the best coalition the smallest, as Riker would have us expect? What if there are more than three parties and several coalitions can form, will Riker's theorem of the minimum winning coalition trump all others? Or are there situations where a larger coalition will outrank the minimum winning coalition? If so, what are conditions making for this situation?

In this paper, we will address these problems of coalitions in situations where vote trading or logrolling is allowed using the terminology and concepts in Coleman's framework. The key concepts to be used in the analysis are

- The control of each actor over an event — e.g. the fraction of the votes he initially controls.
- The interests of each actor in the outcomes of these events — e.g. the difference to him of a positive or negative outcome on each issue.



- The value of the coalition to the collectivity as a whole —
- The value of a coalition for an actor —
- The value of a coalition for the opposition (non-members of the coalition) —

These intuitive concepts are given precise mathematical definitions in the appendix.

We will start with the simplest of situations: Three actors which have interests in three issues over which they share control and can bargain. Next we will expand first the number of issues, and then the number of actors to illustrate some further points.

### 3 Finding the optimal coalition using an extended version of Coleman’s framework

Below, we consider some examples of systems of  $n$  actors tasked with reaching joint decisions on  $m$  issues in a collectivity. We assume that

- the actors can form coalitions within that collectivity;
- that it takes 50+ per cent of the total control (e.g. of votes in a legislature) to form a winning majority;
- that any coalition exceeding at the outset potentially viable.

To avoid trivial single-actor coalitions, each single actor has less than 50% control.

Using our extension of Coleman’s framework sketched above, we calculate the value of each feasible coalition for the individual actors, as well as to the coalition itself, to the opposition, and to the collectivity as a whole. We assume that any actor will prefer coalitions with higher value to himself if given the opportunity to switch.

For each feasible coalition, we check to see if there is an alternative coalition that has a higher value to all actors in that alternative coalition. If there are multiple feasible alternatives that fulfill this criterion, the actors will switch coalition with a probability equal to the relative average gain in value associated with switching. In other words, if switching from coalition  $X$  to coalition  $Y$  yields an average increase in value of 0.2 to the actors in  $Y$ , while switching to coalition  $Z$  has an average increase in value of 0.4 to the actors in  $Z$ , then the probabilities of switching to  $Y$  and  $Z$  are 0.33 and 0.66 respectively.<sup>1</sup>

To introduce the extended framework, we start with some fairly trivial examples to give some intuitions about the framework, then move on to show how control and interest in

---

<sup>1</sup>[*Author note*] “kanskje en footnote som forklarer sammenhengen” [“perhaps a footnote explaining the connection”]

issues interact to generate perhaps counter-intuitive coalitions. Then we move on to some more realistic situations showing how the framework can be applied and generalized.

### 3.1 Example 1: No conflict, no gain

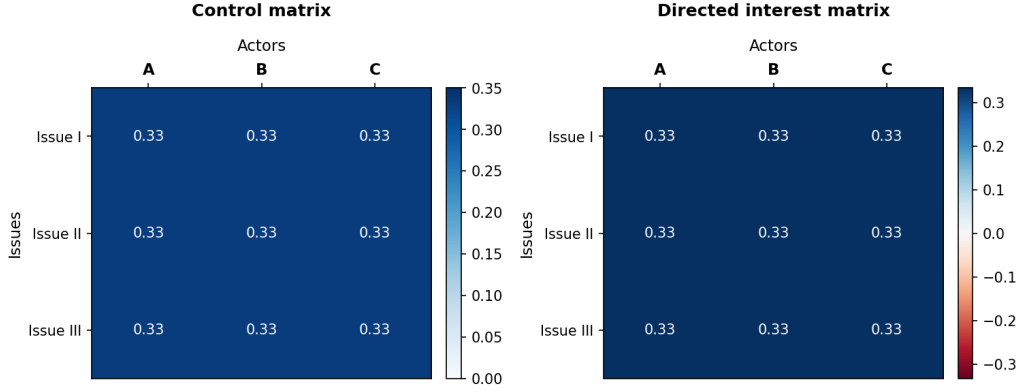


Figure 1: Example 1 — Input matrices. *Left*: Control matrix (uniform  $\frac{1}{3}$  for all actor–issue combinations). *Right*: Directed interest matrix (attitude  $\times$  interest; all actors agree positively on all issues with equal intensity).

In this first example, every actor has equal control over every issue. Furthermore, all agree, positively or negatively as to the preferred outcome of every issue (attitudes are aligned), and all issues are equally important to every actor. This is shown in the uniform control matrix and the directed interest matrix in Figure 1. In Figure 2, the value of the coalition for each actor, for each coalition, for the opposition, and for the collectivity as a whole.

In this and in the figures below, the shaded area identifies possible winning coalitions.

**Example 1 — No conflict, no gain**

| Coalition | Actor A | Actor B | Actor C | Coalition   | Opposition | Collectivity |
|-----------|---------|---------|---------|-------------|------------|--------------|
| B+C       | 0.33    | 0.33    | 0.33    | <b>0.67</b> | 0.33       | 1.00         |
| A+C       | 0.33    | 0.33    | 0.33    | <b>0.67</b> | 0.33       | 1.00         |
| A+B       | 0.33    | 0.33    | 0.33    | <b>0.67</b> | 0.33       | 1.00         |
| A+B+C     | 0.33    | 0.33    | 0.33    | <b>1.00</b> | 0.00       | 1.00         |

Figure 2: Example 1 — Coalition value table.

### Example 1 — coalition map

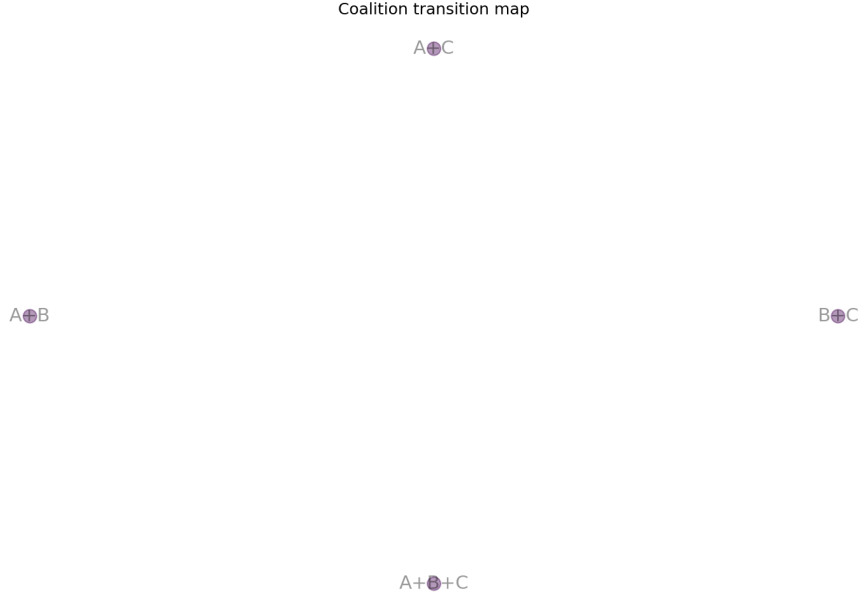


Figure 3: Example 1 — Coalition transition map.

When all actors have aligned attitudes on all issues, all coalitions are equally valuable to every actor. Since the direction of their interests are aligned, no actor has any incentive to break out of the ‘coalition of all’ to form smaller coalitions. That said, it is also irrelevant for the actors if the two other actors decided to break out and form a coalition, because the outcome of the decisions are unaffected by who has the power to decide.

The first conclusion, therefore, is that when all attitudes are aligned and the actors have equal control, coalitions are not necessary.

### 3.2 Example 2: When conflicting attitudes are introduced, being part of a coalition can bring benefits to actors

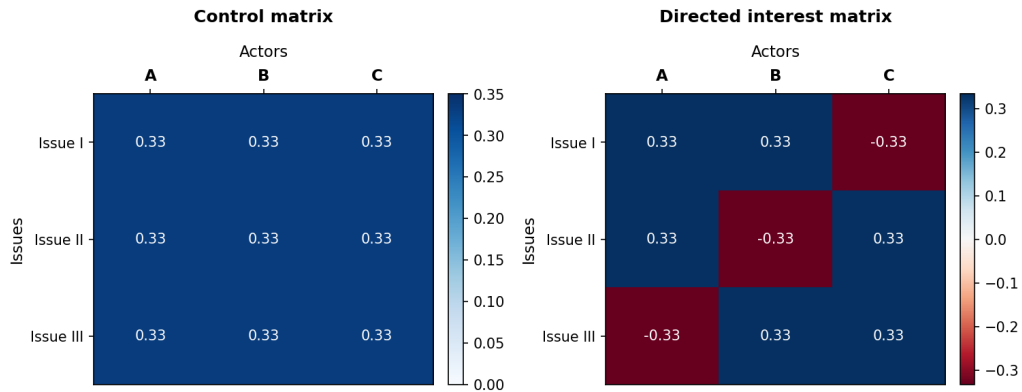


Figure 4: Example 2 — Input matrices. Actors A, B, C each disagree with the other two on exactly one issue (symmetric conflict). Control remains uniform.

### Example 2 — Conflicting attitudes

| Coalition | Actor A | Actor B | Actor C | Coalition   | Opposition | Collectivity |
|-----------|---------|---------|---------|-------------|------------|--------------|
| B+C       | -0.11   | 0.11    | 0.11    | <b>0.22</b> | -0.11      | 0.11         |
| A+C       | 0.11    | -0.11   | 0.11    | <b>0.22</b> | -0.11      | 0.11         |
| A+B       | 0.11    | 0.11    | -0.11   | <b>0.22</b> | -0.11      | 0.11         |
| A+B+C     | 0.04    | 0.04    | 0.04    | <b>0.11</b> | 0.00       | 0.11         |

Figure 5: Example 2 — Coalition value table.

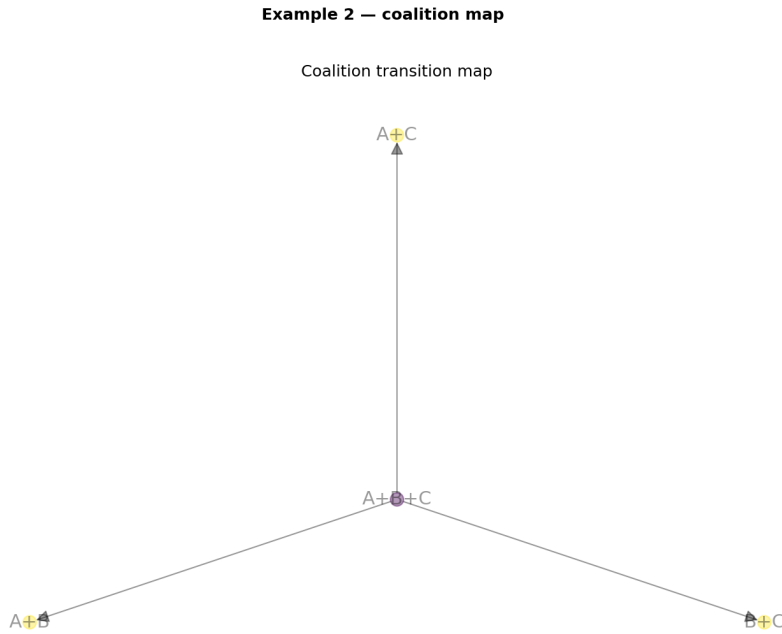


Figure 6: Example 2 — Coalition transition map.

In this example, the actors are no longer in agreement on all issues. In fact, all three actors are in disagreement with the two others on exactly one issue. Save this symmetric change in attitudes, the parameters here are identical to the previous example.

Since the actors no longer agree on all issues, it is no longer optimal for actors to remain in the ‘coalition of all’. However, due to the symmetries of the situation, all two-actor coalitions are equally valuable for all coalition members. Thus, all actors would prefer to be in a two-actor coalition rather than remaining in the coalition of all. But which coalition they join does not matter (as long as they are not left out!).

Note also that in this case it is possible to identify the loss to the opposition or coalition non-member due to the externalities of the coalition members acting jointly in their interest. Moreover, in such a situation the first movers win out — the left-out party will

not be able to induce any of the coalition members to defect: there is no greater gain for any coalition member to be obtained by joining another. First movers would not incur bargaining or transaction costs for no gain.

### 3.3 Example 3: If actors' interests and control are distributed equally across issues, the actors that agree on most issues will form a coalition

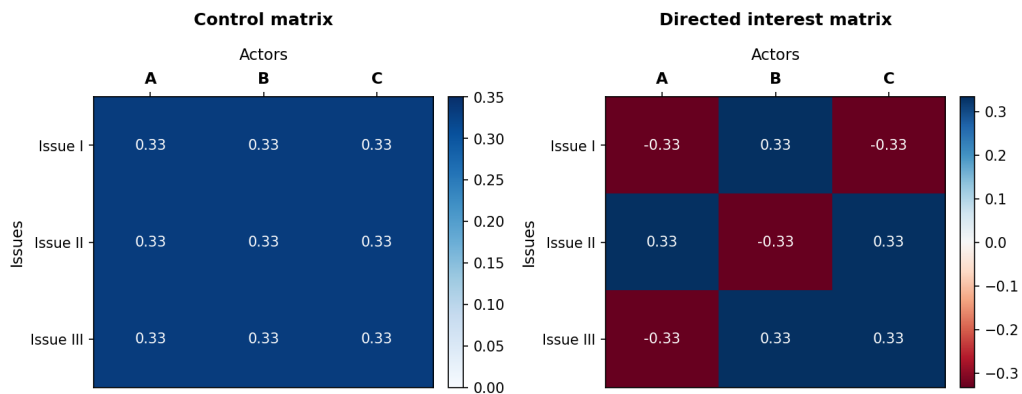


Figure 7: Example 3 — Input matrices. Compared to Example 2, actor A's attitude on Issue I is now negative. A and C now agree on two out of three issues.

#### Example 3 — Agreement on most issues wins

| Coalition | Actor A | Actor B | Actor C | Coalition   | Opposition | Collectivity |
|-----------|---------|---------|---------|-------------|------------|--------------|
| B+C       | -0.11   | 0.11    | 0.11    | <b>0.22</b> | -0.11      | 0.11         |
| A+C       | 0.22    | -0.22   | 0.22    | <b>0.44</b> | -0.22      | 0.22         |
| A+B       | 0.00    | 0.00    | 0.00    | <b>0.00</b> | 0.00       | 0.00         |
| A+B+C     | 0.04    | -0.04   | 0.11    | <b>0.11</b> | 0.00       | 0.11         |

Figure 8: Example 3 — Coalition value table. Coalition A+C (shaded) is the dominant outcome.

### Example 3 — strongest transitions

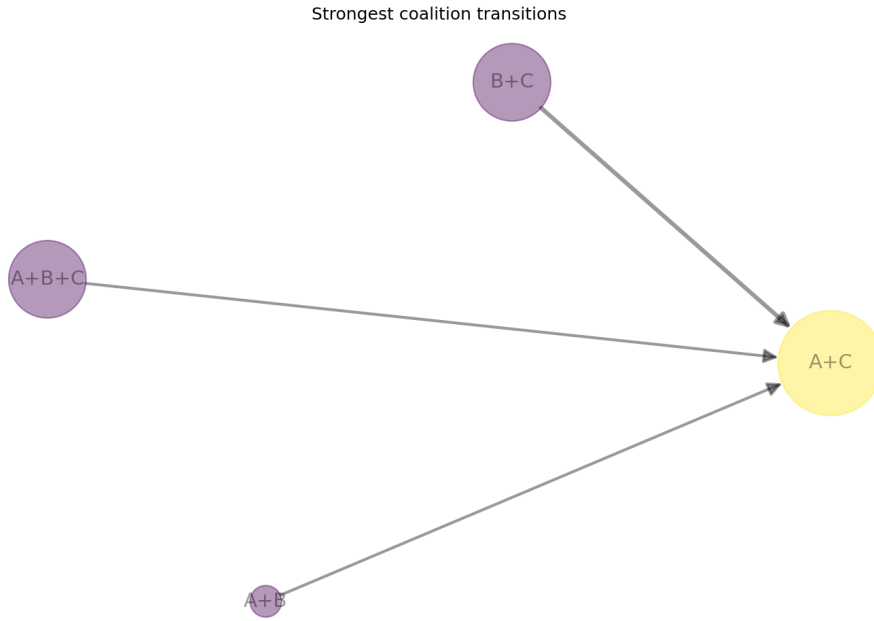


Figure 9: Example 3 — Strongest coalition transitions.

In this example, only a single change has been made in the parameters compared to the previous example: actor A's attitude towards Issue I is now negative. This changes the overall situation: A and B are opposed on every issue, A and C agree on 2 issues (I and II), while B and C only agree on one issue (III).

The outcome of this one change makes the coalition between actors A and C more valuable to them as coalition members than any other feasible coalition they could join. Intuitively, this makes sense: they agree on two of three issues, while all other possible combinations of actors involve more disagreement among the coalition members. Thus, the coalition of actors A and C (shaded) is the one that is likely to realize most of the coalition members' interests. Note that there is a rank order of the coalitions for the collectivity as a whole:  $A+C$  being the greatest (0.22),  $B+C$  in the middle (0.11) while  $A+B$  is the lowest (0.0). Note also that different coalitions have different values to the opposition: The dominant coalition ( $A + C$ ) is the one with the greatest negative externality for the opposition (B).

### 3.4 Example 4: Altering the intensity or relative interest actors have in different issues changes the winning coalition, even when the attitudes and control of actors are unchanged

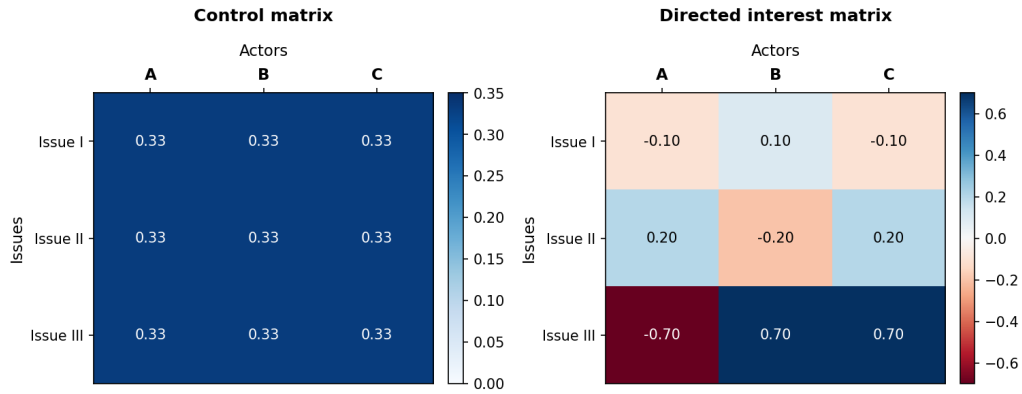


Figure 10: Example 4 — Input matrices. Attitudes and control are identical to Example 3, but interest weights are now skewed: Issue III carries 70% of each actor’s total interest.

**Example 4 — Interest intensity overrides alignment**

| Coalition | Actor A | Actor B | Actor C | Coalition   | Opposition | Collectivity |
|-----------|---------|---------|---------|-------------|------------|--------------|
| B+C       | -0.23   | 0.23    | 0.23    | <b>0.47</b> | -0.23      | 0.23         |
| A+C       | 0.10    | -0.10   | 0.10    | <b>0.20</b> | -0.10      | 0.10         |
| A+B       | 0.00    | 0.00    | 0.00    | <b>0.00</b> | 0.00       | 0.00         |
| A+B+C     | -0.04   | 0.04    | 0.11    | <b>0.11</b> | 0.00       | 0.11         |

Figure 11: Example 4 — Coalition value table. Coalition B+C (shaded) is now dominant.

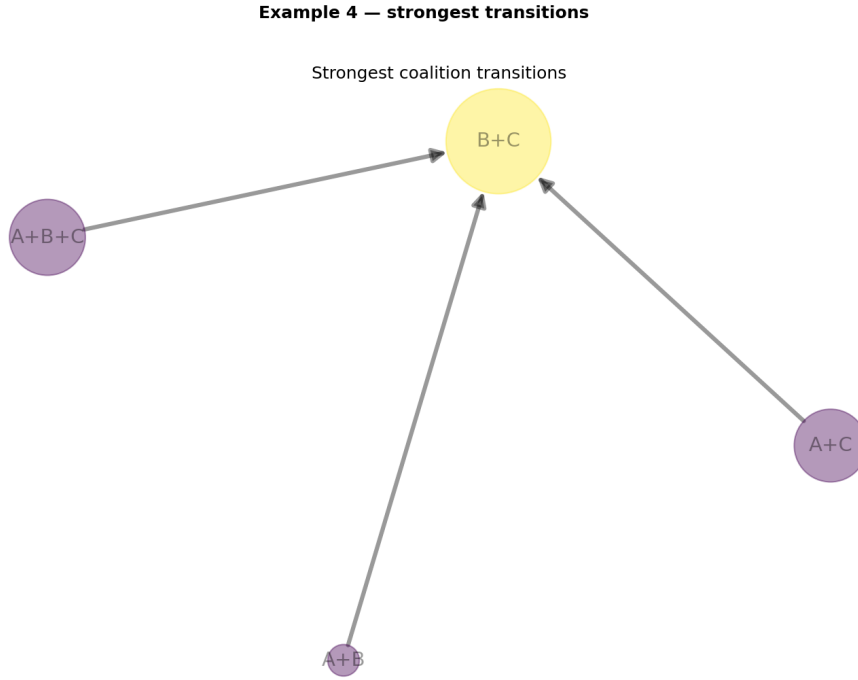


Figure 12: Example 4 — Strongest coalition transitions.

In this example, we keep the control and attitude matrices unchanged from the previous example, but we change the intensities of the actors' interests in the different issues. Intuitively, all actors in this example have strong interests in the outcome of Issue III, while Issues I and II are less important. Now, even though actor B and C only agree (have aligned attitudes) on a single issue (Issue III), they end up forming the winning coalition. This is because they agree on the issue that matters most to them and disagree on issues that are of less importance. Thus, even though A and C have aligned attitudes on Issues I and II, which was enough for them to form the winning coalition in the previous example, the fact that they disagree on the issue that matters most to them is sufficient to change the final value of their coalitions. The change in the intensity of interest in an issue can trump the attitude (positive or negative).

### 3.5 Example 5: In some situations, there is no winning coalition — there is always a better exit option!

Adding more actors to the collective, or more issues they are to make a collective decision on, can make the situation more complex in terms of more diverse or juxtaposed distributions of interest and control. Finding the optimal coalition then becomes less intuitive and some peculiar situations emerge. But it makes it possible to answer such questions as: will a minimum winning coalition emerge as the best alternative? Or can non-minimum coalitions be preferred because they enable the coalition members to realize more of their



interests?

In answering such questions, it is important to note a key difference between Riker's general approach and Coleman's: In the latter not just the size of vote share (weight) of a party (control) is taken into account, but also its directed interest in the outcomes.

#### Author note

HER BØR DET KANSKJE SIES AT ATTITUDE-MATRISEN ER BYGGET INN I INTERESSEMATRISEN. Eller ta med attitude-matrisen, siden det da blir lettere å se konstellasjonene. JA, TA MED ATTITUDEMATRISEN.

[*Translation:* “HERE IT SHOULD PERHAPS BE SAID THAT THE ATTITUDE MATRIX IS EMBEDDED IN THE INTEREST MATRIX. Or include the attitude matrix, since it will then be easier to see the constellations. YES, INCLUDE THE ATTITUDE MATRIX.”]

*Editor's note:* In the revised version, attitude and interest have been merged into the directed interest matrix shown in the figure below; the sign of each cell encodes attitude, the magnitude encodes interest intensity.

To explore the problem, we have in this example randomly generated constellations of control, attitudes and intensities of interest rather than constructing a particular constellation by hand, so to speak.

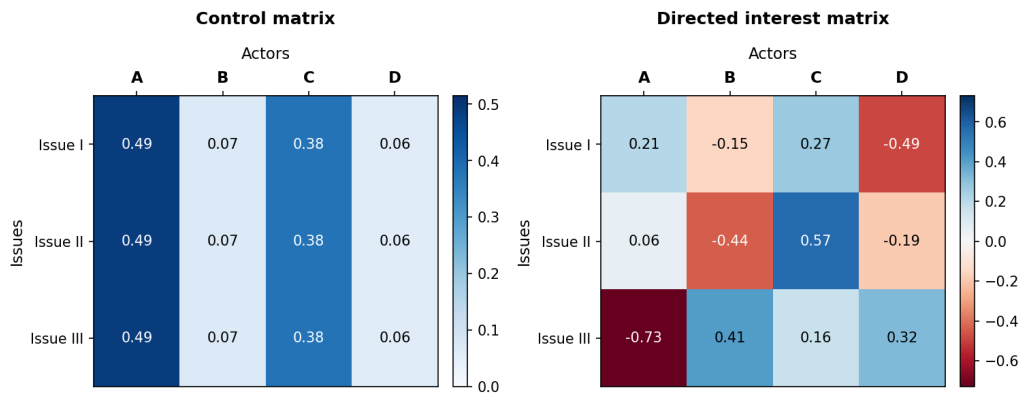


Figure 13: Example 5 — Input matrices for the cycling-loop scenario (4 actors, 3 issues). *Left:* Control matrix (actor A dominates with 49%). *Right:* Directed interest matrix.

**Example 5 — Cycling loop (no stable coalition)**

| Coalition | Actor A | Actor B | Actor C | Actor D | Coalition    | Opposition | Collectivity |
|-----------|---------|---------|---------|---------|--------------|------------|--------------|
| B+C+D     | -0.20   | 0.02    | 0.22    | -0.01   | <b>0.23</b>  | -0.20      | 0.03         |
| A+D       | 0.27    | -0.22   | 0.09    | -0.21   | <b>0.05</b>  | -0.13      | -0.08        |
| A+C       | 0.26    | -0.29   | 0.24    | -0.30   | <b>0.50</b>  | -0.60      | -0.09        |
| A+C+D     | 0.23    | -0.26   | 0.21    | -0.25   | <b>0.18</b>  | -0.26      | -0.07        |
| A+B       | 0.26    | -0.15   | 0.02    | -0.22   | <b>0.11</b>  | -0.20      | -0.09        |
| A+B+D     | 0.21    | -0.10   | -0.03   | -0.14   | <b>-0.03</b> | -0.03      | -0.06        |
| A+B+C     | 0.22    | -0.24   | 0.20    | -0.26   | <b>0.18</b>  | -0.26      | -0.08        |
| A+B+C+D   | 0.19    | -0.21   | 0.17    | -0.21   | <b>-0.06</b> | 0.00       | -0.06        |

Figure 14: Example 5 — Coalition value table. No coalition dominates all others; the system cycles.

**Example 5 — coalition transitions (cycling)**

Coalition transition map

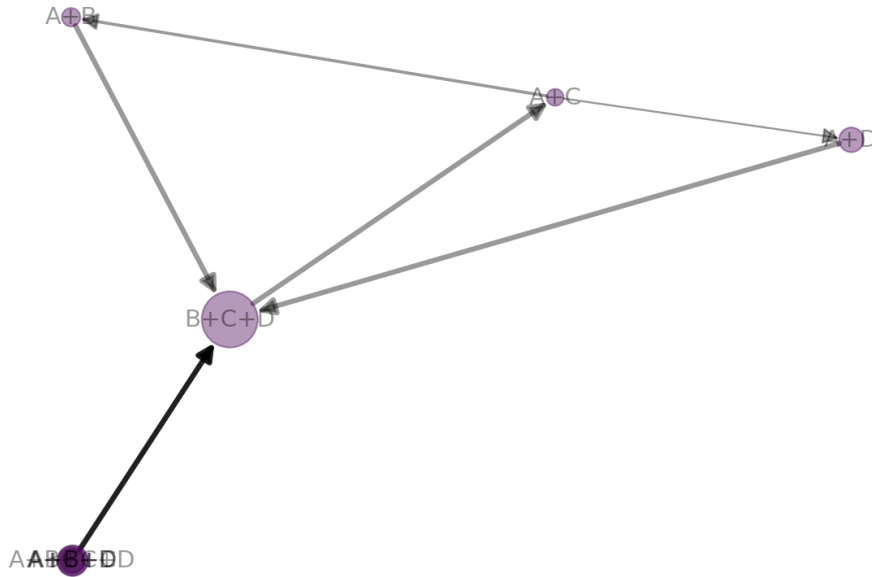


Figure 15: Example 5 — Coalition transition map showing the cycling loop.

### Example 5 — strongest transitions (cycling)

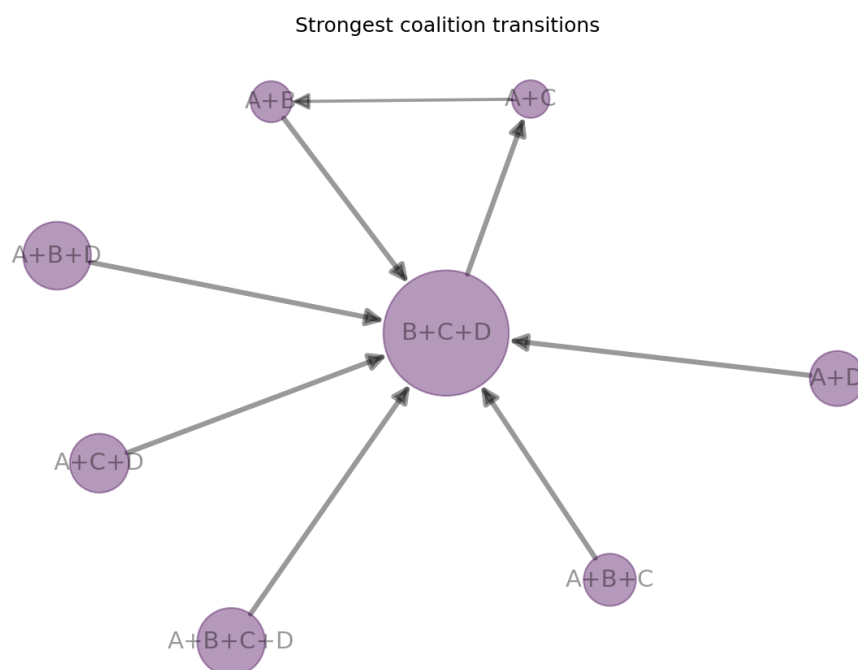


Figure 16: Example 5 — Strongest transitions (cycling).

#### Author note

Vi må bruke bokstaver til å betegne aktører i figuren som i teksten.

[*Translation*: “We must use letters to label actors in the figure as in the text.”]

BØR VEL HA EN TEKST TIL FIGUREN SOM FORKLARER ELEMENTENE I STJERNEDIAGRAMMET.

[*Translation*: “SHOULD HAVE A CAPTION TO THE FIGURE THAT EXPLAINS THE ELEMENTS IN THE STAR DIAGRAM.”]

The resulting dynamics show that situations may occur with no optimal coalition of actors — there is always a better option for at least one of the coalition members. Because of the assumption that any actor will prefer to join a coalition which is more valuable for himself, the collective in this example ends up in an endless loop of negotiations (illustrated by the figure above): one of the coalition members can always be induced to defect given the particular combination of interests and control. In a sense this corresponds to the cyclicity described by Arrow.

#### Author note

HER ER DE TO SPØRSMÅL:

- KAN NOEN KONSTELLASJONER GI INFINITE LOOPS?
- KAN NOEN KONSTELLASJONER GI KOALISJONER SOM ER STØRRE

ENN MINIMUM. VI BØR TROLIG HA ET EKSEMPEL PÅ HVER (KOMMER NEDENFOR I EKSEMPEL 8).

- TROR OGSÅ AT VI BØR VISE I TABELLFORM AT EN VINNENDE KOALISJON IKKE ER MINIMUM.

[*Translation:* “HERE ARE THE TWO QUESTIONS: Can some constellations produce infinite loops? Can some constellations produce coalitions larger than the minimum? We should probably have an example of each (comes below in Example 8). We also believe we should show in table form that a winning coalition is not minimum.”]

In this particular example, the situation arises partly because the two actors B and D have aligned interests on all issues but are not large enough to form a feasible coalition on their own.

### 3.6 Example 6: Breaking the loop

There are probably many ways of breaking out of this kind of loop (for example one could add another issue to be resolved), but the exact solution probably depends on the particular properties of the situation under consideration.

#### Author note

TROR VI BØR TA DETTE SOM ET EGET EKSEMPEL: BREAKING THE LOOP.

[*Translation:* “WE BELIEVE THIS SHOULD BE A SEPARATE EXAMPLE: BREAKING THE LOOP.”]

*Editor’s note:* Example 6 is treated as a separate example in this revised version.

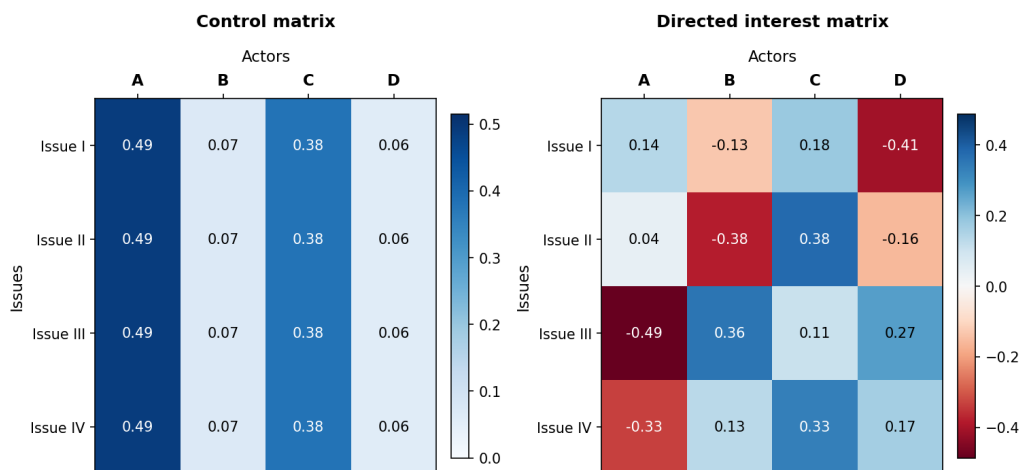


Figure 17: Example 6 — Input matrices after adding Issue IV (where B, C, D broadly agree and A strongly opposes). Values are re-normalised per actor across all four issues.

**Example 6 — Loop broken by adding Issue IV**

| Coalition | Actor A | Actor B | Actor C | Actor D | Coalition    | Opposition | Collectivity |
|-----------|---------|---------|---------|---------|--------------|------------|--------------|
| B+C+D     | -0.19   | 0.05    | 0.18    | 0.05    | <b>0.29</b>  | -0.19      | 0.10         |
| A+D       | 0.20    | -0.16   | -0.04   | -0.16   | <b>0.04</b>  | -0.20      | -0.16        |
| A+C       | 0.14    | -0.20   | 0.11    | -0.19   | <b>0.25</b>  | -0.39      | -0.14        |
| A+C+D     | 0.12    | -0.17   | 0.09    | -0.15   | <b>0.06</b>  | -0.17      | -0.11        |
| A+B       | 0.20    | -0.11   | -0.08   | -0.16   | <b>0.09</b>  | -0.24      | -0.15        |
| A+B+D     | 0.16    | -0.07   | -0.10   | -0.10   | <b>-0.01</b> | -0.10      | -0.11        |
| A+B+C     | 0.11    | -0.15   | 0.08    | -0.16   | <b>0.05</b>  | -0.16      | -0.11        |
| A+B+C+D   | 0.09    | -0.13   | 0.07    | -0.12   | <b>-0.09</b> | 0.00       | -0.09        |

Figure 18: Example 6 — Coalition value table. Coalition B+C+D now dominates.

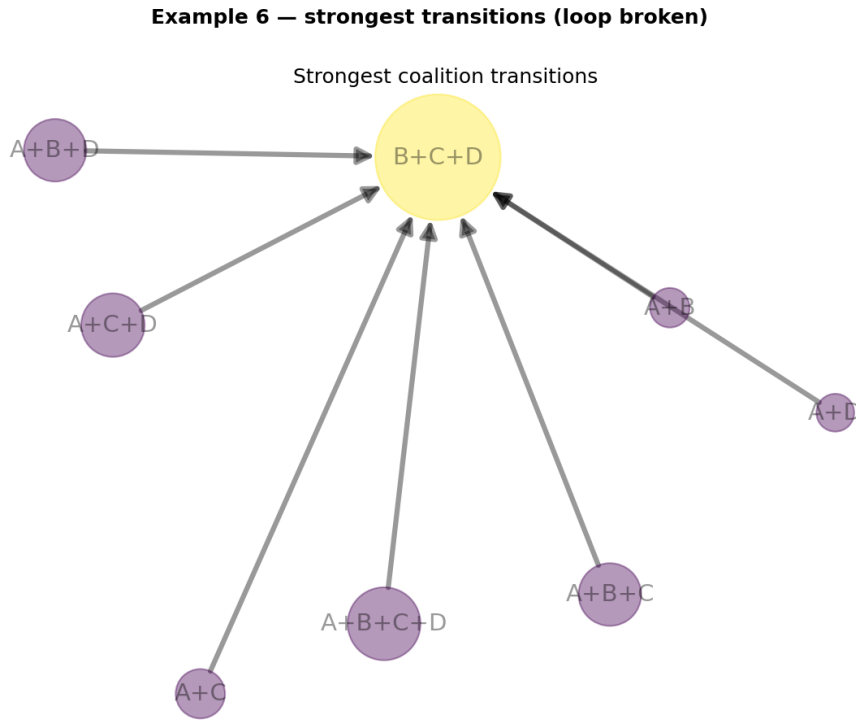


Figure 19: Example 6 — Strongest transitions; the loop is broken.

### 3.7 Example 7: Path dependence — the optimal coalition may depend on the starting point of negotiations

#### Author note

FORKLAR FARGENE I KOLONNEN FOR COALITION MEMBERS NEDENFOR.  
 [Translation: “EXPLAIN THE COLOURS IN THE COALITION MEMBERS COLUMN BELOW.”]

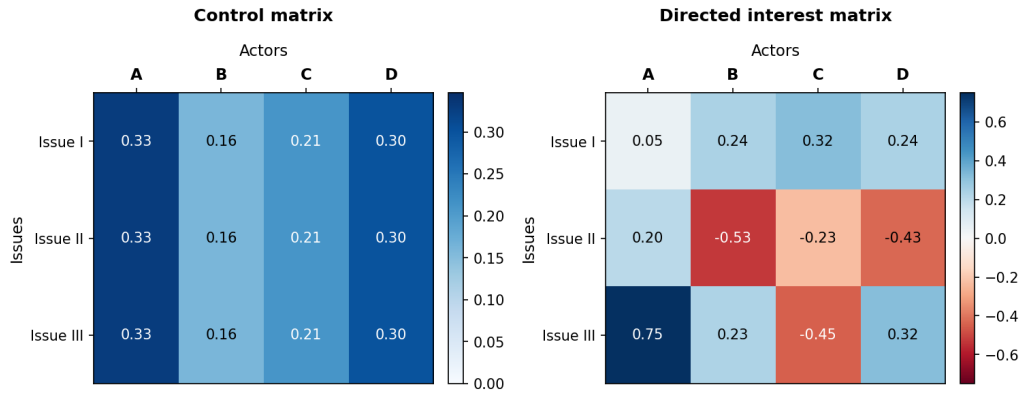


Figure 20: Example 7 — Input matrices (path dependence scenario, 4 actors, 3 issues).

#### Example 7 — Path dependence

| Coalition | Actor A | Actor B | Actor C | Actor D | Coalition   | Opposition | Collectivity |
|-----------|---------|---------|---------|---------|-------------|------------|--------------|
| C+D       | -0.05   | 0.26    | 0.18    | 0.23    | <b>0.41</b> | 0.21       | 0.62         |
| B+C+D     | -0.01   | 0.27    | 0.16    | 0.24    | <b>0.67</b> | -0.01      | 0.66         |
| A+D       | 0.24    | 0.21    | -0.02   | 0.24    | <b>0.48</b> | 0.20       | 0.68         |
| A+C       | 0.14    | 0.09    | 0.03    | 0.11    | <b>0.17</b> | 0.19       | 0.36         |
| A+C+D     | 0.13    | 0.20    | 0.06    | 0.21    | <b>0.39</b> | 0.20       | 0.60         |
| A+B+D     | 0.23    | 0.25    | -0.00   | 0.27    | <b>0.75</b> | -0.00      | 0.74         |
| A+B+C     | 0.12    | 0.18    | 0.06    | 0.18    | <b>0.35</b> | 0.18       | 0.54         |
| A+B+C+D   | 0.13    | 0.23    | 0.06    | 0.23    | <b>0.65</b> | 0.00       | 0.65         |

Figure 21: Example 7 — Coalition value table. Two coalitions (B+C+D and A+B+D) are both winning; which one prevails depends on the starting point.

**Example 7 — coalition map (multiple sinks = path dependence)**

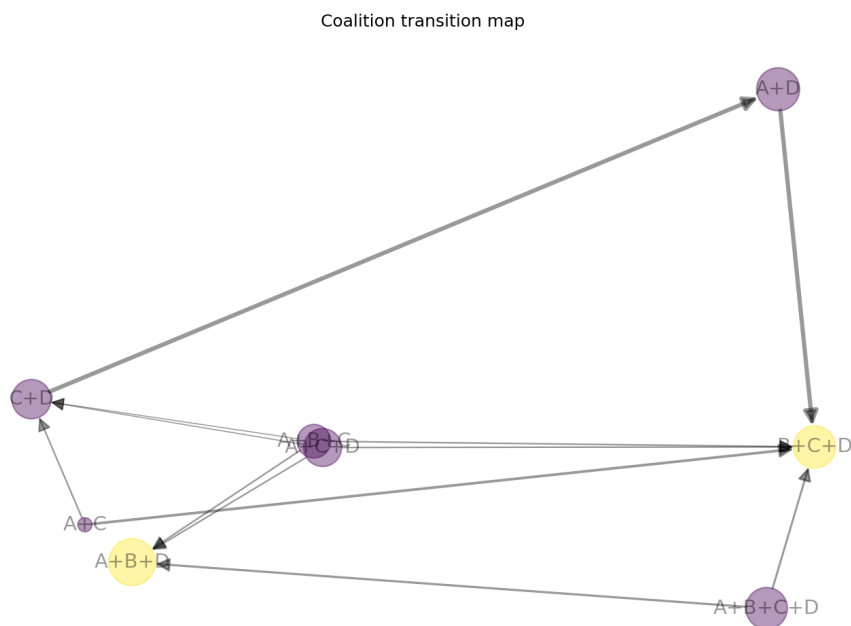


Figure 22: Example 7 — Coalition map showing multiple sinks (path dependence).

**Example 7 — strongest transitions**

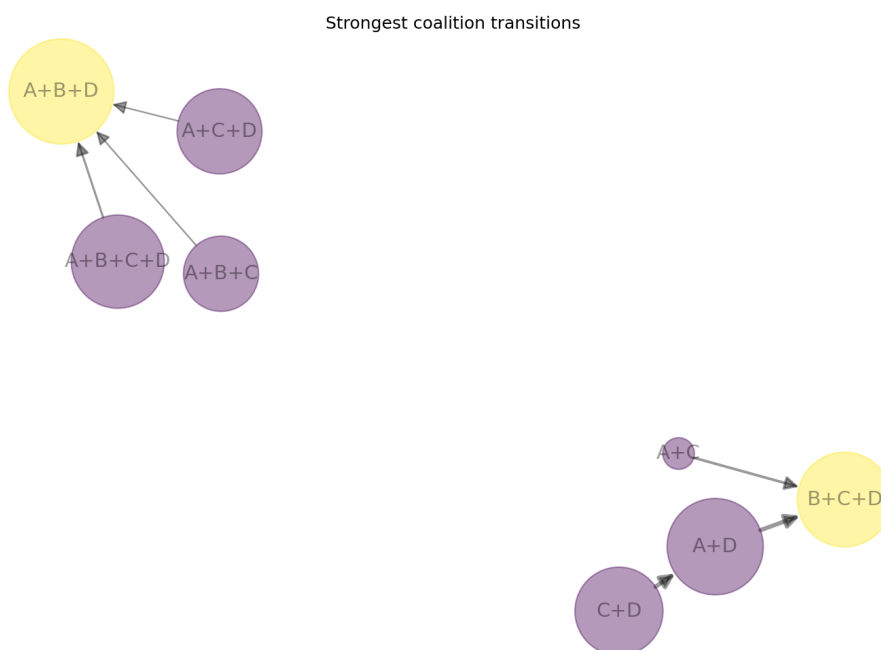


Figure 23: Example 7 — Strongest transitions.

**Author note**

IGJEN. HVA FIGUREN FORTELLER BØR FORKLARES — KANSKJE EGEN FIGUR VED SIDEN AV TABELLENE.

[*Translation:* “AGAIN. WHAT THE FIGURE CONVEYS SHOULD BE EXPLAINED — PERHAPS A SEPARATE FIGURE ALONGSIDE THE TABLES.”]

Another example of a simulated situation in which the actors’ control and intensity of their interests were randomly generated is shown above. The result shown is quite common in simulations with more than three actors: there is more than one winning coalition.

More than that, the coalition that ends up winning in a given situation may depend on the starting point of negotiation. In this example, if actors C and D

#### Author note

HER BØR EN FÅ SAMME NOTASJON I FIGURENE [C=2, D=3], EVENTUELT I EN HÅNDEGNET VERSJON.

[*Translation:* “THE FIGURES SHOULD USE THE SAME NOTATION [C=2, D=3], OR ALTERNATIVELY IN A HAND-DRAWN VERSION.”]

*Editor’s note:* Actors are now labelled A–D consistently in all figures.

start out trying to form a coalition alone, the winning coalition in the collective will eventually include actor B as well. However, if the initial attempt included the same two actors together with actor A, the winning coalition will be with actors A, B and D, according to this model.

### 3.8 Example 8: In a situation with a known distribution of control among actors, the winning coalition depends on the actors’ particular attitudes and interests — and it is not always minimal

In a Wikipedia article presenting Riker’s Theory of political coalitions, a numerical example is given to illustrate his theorem of minimum winning coalitions, i.e. the rationale for why actors tend to create coalitions that are just large enough to ensure winning and no larger. In that example there are five actors or parties, with control proportional to their size, as follows: A: 0.05, B: 0.40, C: 0.26, D: 0.25, E: 0.04.

#### Author note

KANSKJE LAGE EN KONTROLL-MATRISE FOR Å HOLDE OSS TIL EN EN-SARTET PRESENTASJONSMÅTE.

[*Translation:* “PERHAPS CREATE A CONTROL MATRIX TO MAINTAIN A UNIFORM PRESENTATION STYLE.”]

*Editor’s note:* The control matrix is shown in Figure 24.



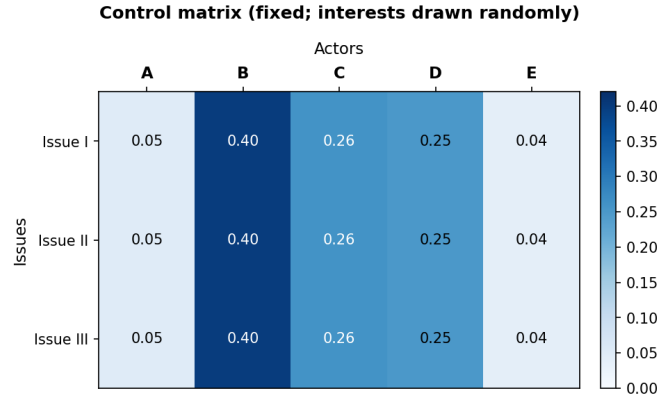


Figure 24: Example 8 — Control matrix (fixed across all 1 000 simulations; interest matrices are drawn randomly).

However, in this model no issues are specified, neither the actors' attitudes nor intensity of interest in them — only the numerical weight or amount of control is included in the model. In other words, there is no Attitude Matrix or Interest Matrix.

Clearly, there are many ways to form potentially majority coalitions in this collective.

**Author note**

HER TROR JEG VI BØR LISTE DE KOALISJONENE I EN TABELL SOM ER MULIG RENT NUMERISK, A LA DETTE:

[*Translation:* “HERE I THINK WE SHOULD LIST THE COALITIONS IN A TABLE THAT IS POSSIBLE NUMERICALLY, ALONG THESE LINES:”]

Table 1 lists all majority coalitions by vote share.

Table 1: Example 8 — Possible majority coalitions by numerical vote share.

| Coalition         | Vote share |
|-------------------|------------|
| A + B + C         | 0.71       |
| B + C             | 0.66       |
| B + D             | 0.65       |
| C + D             | 0.51       |
| B + C + D         | 0.91       |
| B + C + E         | 0.70       |
| C + D + E         | 0.55       |
| A + B + C + D + E | 1.00       |

In Riker's thinking, the minimal viable coalition would emerge as winning, which he would argue is the most valuable to the actors. It would maximize the control of the coalition members and spoils would have to be divided among fewer actors. In the present example,

the coalition between C and D with  $(0.26+0.25 =) 0.51$  of the votes is the minimal winning coalition.

However, if we take into account that the actors not only have different control over the outcome of issues (represented by the percentage of the votes), but different attitudes as well as intensities of interests, it is not a priori clear which coalition will emerge.

To show this, we simulated 1 000 different situations with randomly generated interest and attitude matrices, keeping the control matrix constant. We then recorded the winning coalitions in each simulation and counted the total number of wins for each coalition across the 1 000 simulations.

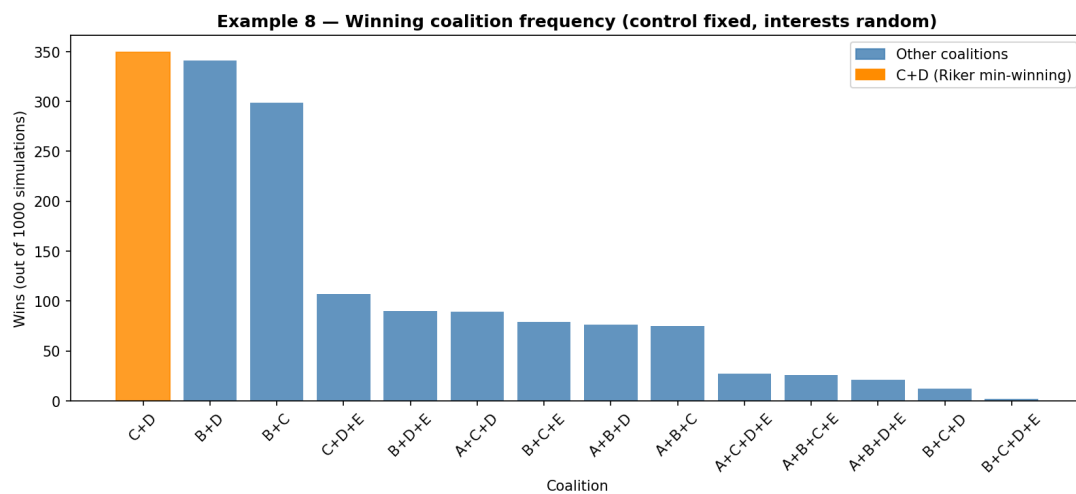


Figure 25: Example 8 — Winning coalition frequencies across 1 000 simulations with fixed control and random directed interest matrices. The Riker minimum-winning coalition (C+D, orange) wins in fewer than 35% of cases.

#### Author note

BRUK A, B, C, D, E I STEDET FOR 1, 2, 3, 4, 5 FOR KONSISTENS.

[*Translation*: “USE A, B, C, D, E INSTEAD OF 1, 2, 3, 4, 5 FOR CONSISTENCY.”]

*Editor’s note*: All figures now use letter labels A–E.

From Figure 25, it appears that the coalition between actors C and D is in fact the most frequent winner, yet the minimum coalition is winning in less than half of the simulated situations. The upshot is that a model of coalition formation has to include not only the control of the actors, but their interests and attitudes as well to better predict outcomes in real-life situations.

From these simulations it is also clear that two coalition parties may sometimes be better served by bringing in a third party to better realize their interests, e.g. by C and D adding a third actor.

#### Author note

ER DET EN AV DE MINSTE PARTENE DE BRINGER INN?

[*Translation: “IS IT ONE OF THE SMALLEST PARTIES THEY BRING IN?”*]

**3.9 Example 9: If the attitudes and interests are held constant, the winning coalition depends on the distribution of control within the collective — and again the winning coalition is not always minimal**

**Directed interest matrix (fixed; control drawn randomly)**

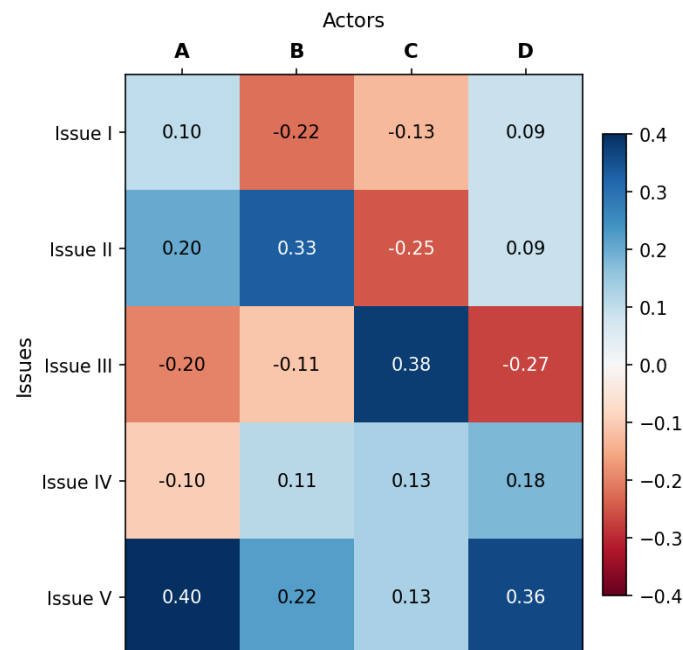


Figure 26: Example 9 — Directed interest matrix (fixed across all 1 000 simulations; control matrices are drawn randomly).

Running 1 000 simulations holding the interests and attitudes of four actors constant, but changing their amount of control over issues for every new simulation, demonstrates that the distribution of control among the actors within the collective impacts on the choice of an optimal coalition. However, it seems that it may have less of an impact than altering the attitudes and interests.

#### Author note

HVA ER KRITERIET FOR DENNE KONKLUSJONEN? OG I HVILKEN GRAD AVHENGER DENNE KONKLUSJONEN AV DE SPESIFIKKE ATTITUDE/INTEREST MATRISENE SOM ER BRUKT?

[*Translation:* “WHAT IS THE CRITERION FOR THIS CONCLUSION? AND TO WHAT EXTENT DOES THIS CONCLUSION DEPEND ON THE SPECIFIC ATTITUDE/INTEREST MATRICES USED?”]

Using the directed interest matrix shown in Figure 26, we see that the coalition of actors A and B come out as winning a majority of the simulations, even though it is only one out of three pairs of actors with three (out of five) aligned attitudes. However, it seems that the fact that they are the ones that agree on the most important issues, so that it is the most valuable coalition in any instance where they together form a feasible coalition.

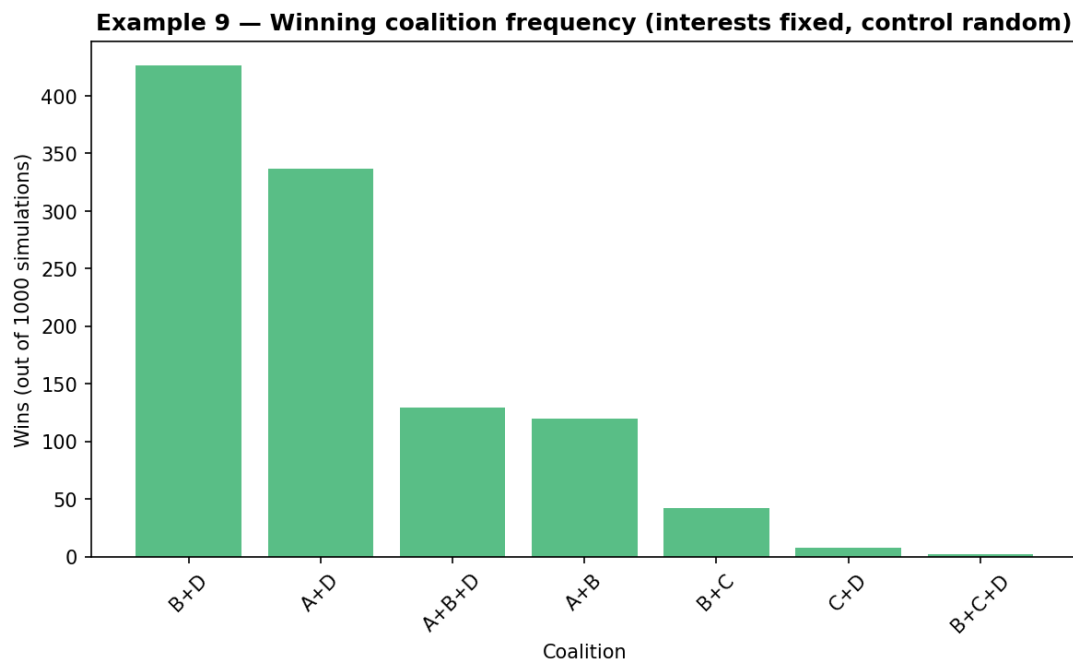


Figure 27: Example 9 — Winning coalition frequencies across 1 000 simulations with fixed directed interest and random control matrices.

#### Author note

##### HVA MED BRIDGING —

- INTRODUSERE EN NY SAK FOR Å GI GRUNNLAG FOR EN ANNEN KOALISJON
- HVA KJENNETEGNER SAKER SOM KAN BRUKES TIL BROBYGGING (IN TERMS OF ATTITUDE AND INTENSITY OF INTEREST — ER DET BEST Å INTRODUSERE EN AKTØR SOM ER NOE UENIG MED DEM SOM FORSØKER Å ETABLERE EN KOALISJON, MEN PÅ ULIK MÅTE? ALTSÅ LITT ALLIERT MED BEGGE, LITT MOTSATT AV BEGGE?
- GJELDER SAMME LOGIKK FOR Å INTRODUSERE EN NY AKTØR SOM FOR Å INTRODUSERE EN NY SAK?

[*Translation:* “WHAT ABOUT BRIDGING — Introduce a new issue to provide a

basis for another coalition. What characterises issues that can be used for bridge-building (in terms of attitude and intensity of interest)? Is it best to introduce an actor who somewhat disagrees with those trying to establish a coalition, but in different ways? That is, partly allied with both, somewhat opposed to both? Does the same logic apply to introducing a new actor as to introducing a new issue?”]

## 4 Preliminary conclusions and questions for further research

- The size of the potential coalition parties alone is not sufficient to predict which coalitions may form. Attitude and interest must be included.
- When they are included, it is fairly easy to identify situations

### Author note

[*Unfinished*] “...at noen-minimum, loops, noen-minimum winning etc”  
[“...where some are non-minimum, loops, non-minimum winning, etc.”]

## 5 Discussion of strengths and weaknesses of the model

### Strengths:

- Simple
- Intuitive
- Linked to economic theory
- Takes attitude and interest into account — this is a realistic assumption — just including control is unrealistic in real life

### Weaknesses:

[*To be completed*]

### Suggestions for further research

- We have primarily provided and analyzed examples. These are interesting.
- Is it possible to identify more general conclusions, e.g. in the form of theorems — e.g. about when a minimum coalition is not the best?
- Can one formulate maxims for what potential coalition parties should do in bargaining?

#### Author note — Ideas for further research (section outline)

We may also ask whether the sequence in which different coalitions are attempted will affect which one in the end is established. In other words: if there are several possible feasible coalitions, will the one that in fact is formed depend on which proto-coalition is first tested out, so that this initial coalition so to speak locks the system to the coalition that is finally realized? Or will the initial bargaining be a kind of round-robin where the finally established coalition will be the same, irrespective of starting point? Or is there a best path towards a stable solution?

If we find situations where a coalition is not able to form, due to the particular constellation of actors' interests and control, what could break a logjam? Is a solution — even a solution for protracted cycles of coalition-formation — to bring in a new issue or a new actor that, so to speak, could serve as a bridge between the original parties?

#### Detailed outline of examples and open questions (author working notes):

1. No conflict, no gain: all or no coalitions are equally beneficial. It is a model of overall harmony.
2. When there are opposed interests or conflicts in attitudes, benefits of coalitions can be reaped. Yet when control is distributed equally and interests symmetrically opposed any one coalition can be established. (Perhaps make a few expansions of actors or issues to buttress this.)
3. When interests and control are equal, actors with aligned attitudes will form a coalition. The plot begins to thicken.
4. Intensity of interest may trump alignment of attitudes.
  - Perhaps a possible parallel finding here:
    - Those with strong but unusual preferences may be more willing to join a coalition — as long as they get their *hjertesak* (pet issue) through, they can accept anything on the others. Does this mean that parties with one overriding but specific concern are most likely coalition parties?
    - Does this mean that coalitions are most easily formed when the actors have different/complementary overriding concerns? This is the logic of “Den Solbergske parlamentarisme”: H gets tax cuts, KrF gets abortion, V gets Senja/Lofoten etc. Is the formula that parties with strong but complementary interests on issues easily can carry the day? But also that if they clash on a specific issue, the coalition is in danger (V and FrP mht bompenger/tolls).
5. Rotating coalitions: some constellations of interest and control means that a member of a (proto)coalition may be enticed out. ‘The grass is greener on the

other side.’

6. (What can break the loop and end rotation?)

- Topic for further research: what can break the infinite circle of bargaining, and in real life cumulative transaction costs?
- Question: Bring in a new actor? Bring in a new issue?
- Coleman’s model is one of logrolling. What when there is no coalition able to form: Belgium had a general election on 13 June 2010 — it took 541 days, until 6 December 2011, to form a coalition; in Sweden it took several months to negotiate the parliamentary basis for a new government in 2018–2019. So what does it take to break a logjam?
- If an additional issue redistributes control among the actors, a different coalition may be feasible.
- Another issue may redistribute the relative interests of the potential coalition partners across issues.

7. Path dependence — who’s first may conquer the stage.

8. The best coalition not necessarily the minimum winning.

- What characterises the constellation of interest/control where the best/most attractive coalition is not the minimum winning?

9. If the attitudes and interests are held constant, the winning coalition depends on the distribution of control within the collective — and again the winning coalition is not always minimal.

- What can we say in general about what gives rise to non-minimum winning coalitions?

**Internal editorial correspondence (not for publication)**

*The following is an internal email/note found at the end of the original manuscript file. Reproduced here for completeness.*

Hei Bjørn,

Da har jeg — med forsinkelser — gått gjennom manuset, og har kuttet og endret. Det er stadig en del å gjøre der men jeg syns også stadig at vi har interessante konklusjoner pluss suggestions for further research.

Om det videre arbeid.

Jeg har forutsatt at vi presenterer matematikken i modellen i et appendiks. Der bør vi også være eksplisitte på hvilke nye begreper vi innfører innen rammen av Colemans modell, men som går ut over de begrep han definerte — det er i seg selv et bidrag. Så til simuleringene, der jeg nedenfor har satt opp rekkefølgen og lagt inn endel kommentarer for videre arbeid.

Innledningsvis sier vi at vi begynner med den enkleste situasjon, including only three

actors, og så bruker modellen til å legge på komplikasjoner.

## A Mathematical formulation

*[Placeholder — formal mathematical definitions of the interest matrix, control matrix, equilibrium conditions, and derived coalition-value quantities to be written here.]*